MECHANICAL PROPERTIES OF FLIP-CHIP BONDING STRUCTURES FOR MICRO-LED DEVICES: CU-CU BONDING WITH PASSIVATION LAYER AND INDIUM BUMPS BONDING

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ABSTRACT

A traditional bonding structure with indium bumps and a new bonding structure with passivation layer using Cu bumps were simulated to bond Micro-LED and substrate. Anand model and bilinear isotropic hardening model were used to characterize the viscoplastic behavior of In and the plastic behavior of composite structure simplified from the combination of Cu and Au, respectively. Mechanical properties of two kinds of micro-display devices with different bonding structures were analyzed by the ANSYS finite element method and the thermo-mechanical coupling field. The deformation and stress of the micro-display device using Cu-Cu bonding structure with passivation layer are reduced by 19.5% and 59.1% compared with the device bonded by In bumps.

INTRODUCTION

In recent years, Micro-LED devices had become a research hotspot in the field of display and visible light communication [1-3]. As the key technology to realize heterointegration, indium bump was widely used to bond Micro-LED chips and Si substrates[4]. However, as the pixel pitch decreased, the extrusion and overflow of indium produced during bonding process may lead to interconnection of the adjacent solder joints and the failure of the devices[5].

Cu-Cu direct bonding was regarded as the more promising integration technology for three dimensional integrated circuit (3DIC). The interconnection between solder joints could be avoided because of the high melting point of Cu[6]. However, the required high vacuum environment to prevent oxidation of Cu and high temperature for Cu-Cu direct bonding limit the application of Cu-Cu bonding in heterointegration[7,8].

A new bonding structure with passivation layer on Cu-Cu bonding interface was proposed to prevent oxidation and promote the diffusion of Cu[9]. Inert metals such as gold, silver or palladium were considered as passivation layer[10-12]. The technological process of inserting passivation layer on Cu-Cu bonding interface is shown in Figure 1[12]. Ti, Cu and passivation layer were deposited on silicon wafer by physical vapor deposition. During the thermal compression bonding process, Cu atoms diffuse to the bonding interface through the passivation layer to complete the Cu-Cu bonding.

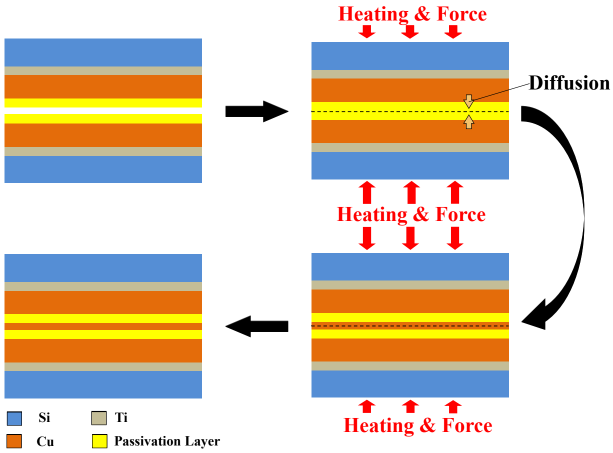

Figure 1: Schematic of Cu-Cu bonding with passivation layer.

Cu-Cu bonding structure with passivation layer was introduced to Micro-LED device to improve mechanical properties and prevent bridging failure of solder joints.

Mechanical properties of the micro-display devices with Cu-Cu bonding structure with passivation layer and micro-display device with indium bonding structure were analyzed by the ANSYS finite element method.

3D FINITE ELEMENT MODEL

Model & Parameters

Structure of the Micro-LED array bonded on the Si substrate through different bonding structures were used for ANSYS analysis. 1/4 model shown in Figure 2 was selected as the object of simulation because of the excellent symmetry of the overall structure. The micro-display device using In bonding structure was named Device A and the micro-display device using

Cu-Cu bonding structure with passivation layer was named Device B.

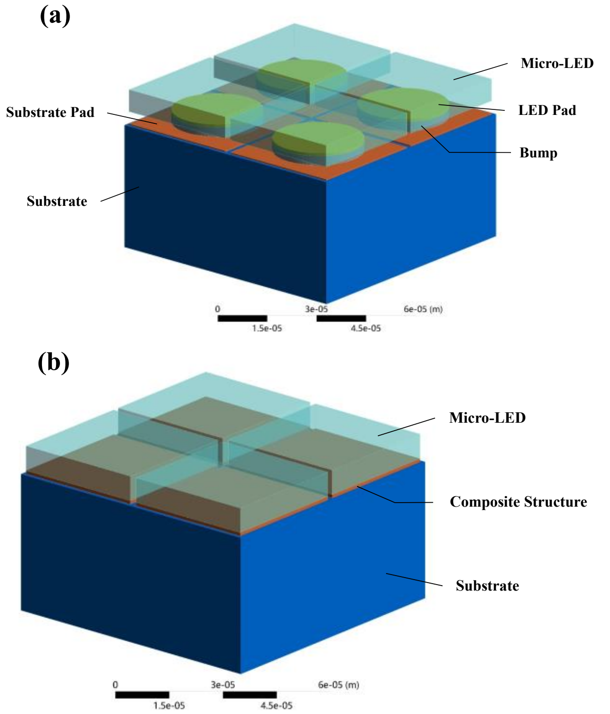

Figure 2: The 1/4 3D model of micro-display devices. (a) Device A (b) Device B

Considering the accuracy of the results and the simulation time as short as possible, the meshing size of 2μm was selected as the simulation parameter.

The details of materials and dimensions for the two models are shown in Table Ⅰ. The composite structure is simplified from the combination of Cu and Au. The thermodynamic parameters of composite structure are related to the thicknesses and the parameters of Cu and Au. The fitting formula of composite parameters is as follows:

$$X_{co} = \frac{H_A}{H_P} X_A + \frac{H_C}{H_P} X_C \quad (1)$$

X_{co}, X_A and X_C are the thermodynamic parameters of composite structure, Au and Cu. H_P, H_A and H_C are the thicknesses of composite structure, Au layer and Cu layer, respectively.

TABLE I DIMENSION OF MODEL PARTS

Parts	Material	Dimension/μm
Micro-LED	GaN	38×38×7.2
LED Pad	Au	Φ20×1
Substrate Pad	Au	38×38×0.4
Bump	In	Φ20×3
Substrate	Si	160×160×40
composite structure	Cu&Au	38×38×0.86

Material Properties

The melting points of In, Au and Cu are 156 ℃, 1064 ℃ and 1083 ℃. The viscoplastic behavior of In and the plastic behavior of Au and Cu were considered in the analysis of the mechanical properties of micro-display devices. Anand constitutive model was used to characterize the viscoplastic behavior of In. Table Ⅱ shows the Anand model parameters of In[13]. The bilinear isotropic hardening model was used to describe the plastic behaviors of Au, Cu and the composite structure. The required parameters of this model are shown in Table Ⅲ, and other linear parameters are shown in Table Ⅳ[2,13-15].

TABLE II ANAND MODEL CONSTANTS OF INDIUM

Description	Indium
So(MPa)	28.3
Q/R(K)	9369.7
A(SEC^-1)	2.33×108
ξ	49.97
m	0.3
ho(MPa)	500
Ŝ(MPa)	28.3
n	0.005
a	1

TABLE III PARAMETERS OF BILINEAR MATERIALS

Material	Yield Strength (MPa)	Tangent Modulus (MPa)
Cu	172.3	517.1
Au	100	5500
composite structure	167.8	943.8

TABLE IV LINEAR PARAMETERS OF MATERIALS

Material	E (GPa)	Poisson's ratio	CTE (10^{-6}/K)	ρ (kg/m^3)
GaN	181	0.35	4.7	6070
In	10.6	0.45	33	7310
Si	190	0.28	2.6	2330
Cu	171	0.35	14.3	8900
Au	79.5	0.34	14	19320
composite structure	165	0.351	14.35	9728

TABLE V THERMODYNAMIC PARAMETERS OF MATERIALS

Material	Thermal conductivity (W/m·K)	Specific heat capacity (J/kg·K)
GaN	175	54.7
In	81.8	233
Si	150	729
Au	317	129
Cu	398	385
composite structure	394	367.8

Figure 3 shows the stress-strain curves of the three bilinear materials drawn according to the mechanical parameters shown in Table Ⅲ. Most of material in the composite structure is Cu, so the curve of the fitted composite structure is similar to that of Cu. Table Ⅴ shows the thermodynamic parameters of these materials[9].

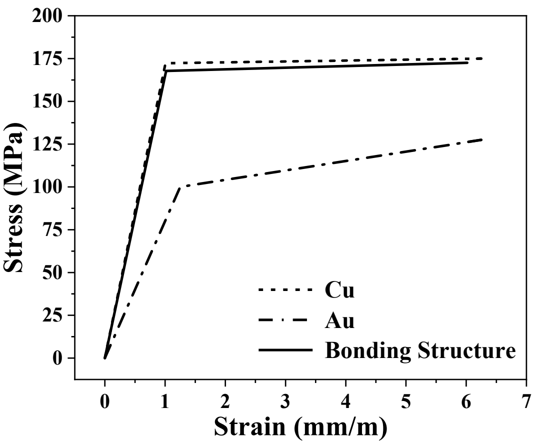

Figure 3: Stress-strain curves of bilinear materials

Loading and boundary conditions

The thermo-mechanical coupling field was used to analyze the temperature profile, deformation and stress distribution of the model. The external ambient temperature was 20 ℃. The heat generation rate of Micro-LED was set to 3.2×10^{8} W/m^{3} to simulate the heat generation of Micro-LED. The convection coefficient of the lower surface of the Si substrate was 10 W/(m^{2}·K) and the fixed support was added to the center of upper surface of the Si substrate[9].

RESULTS AND DISCUSSION

With the increase of temperature for Micro-LED, the heat emitted from the surface of the Si substrate to the environment increases gradually. The temperature of the whole structure reaches a fixed value according to a state of dynamic equilibrium of the device finally.

Figure 4 shows the deformation distribution of the two kinds of devices. During the heating process, the mismatch of the thermal expansion coefficient of the materials leads to the warping deformation of the devices. The deformation increases gradually from the center to the periphery, and it is proportional to the distance from the center of device.

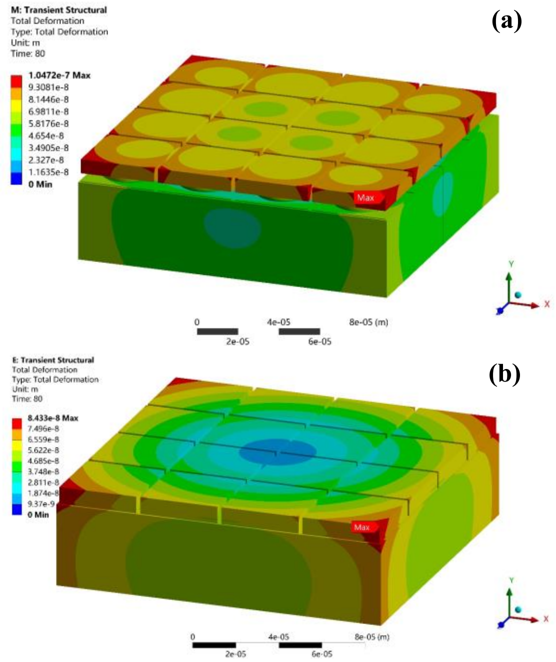

Figure 4: The deformation of micro-display devices. (a) Device A (b) Device B.

Figure 5 shows the stress distribution of two kinds of devices. Figure 6 is the magnification of the local details of Figure 5. The maximum stress concentration point is located at the contact interface between Si Substrate and bonding structure.

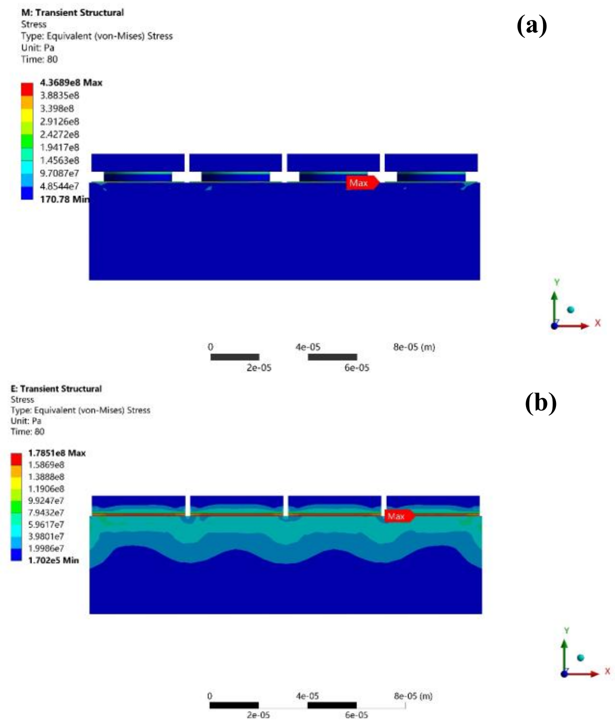

Figure 5: The stress of micro-display devices. (a) Device A (b) Device B.

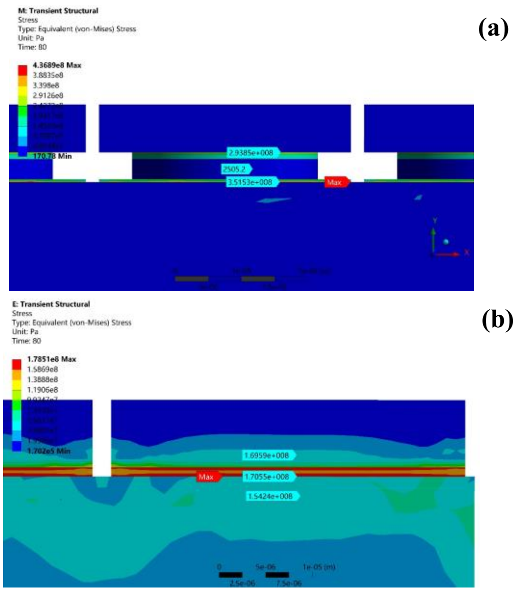

Figure 6: Schematic diagram of maximum stress area. (a) Device A (b) Device B

The steady-state temperature, strain and stress of micro-display devices with different bonding structures are shown in Figure 7. The steady-state temperatures of devices are both 227.93 ℃, which indicates that their heat dissipation performance is similar.

The maximum deformation of Device A is 104.72 nm, and that of Device B is 84.33 nm. In addition, stress of micro-display device reduces from 436.89 MPa to 178.51 MPa by using bonding structure with passivation layer to take place of indium bumps bonding structure.

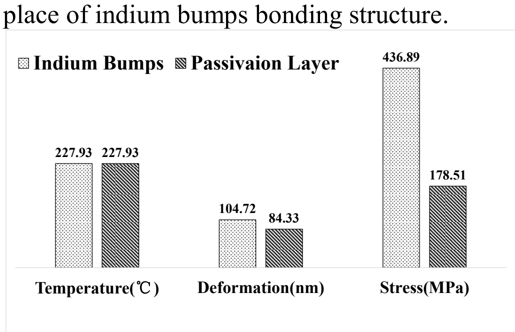

Figure 7: Properties of micro-display devices with different bonding structures.

CONCLUSIONS

The mechanical properties of two kinds of Micro-LED devices bonded by indium bumps and Cu-Cu bumps with passivation layer were investigated by ANSYS method.

The mechanical properties of Micro-LED device is improved effectively by using of Cu-Cu bonding structure with passivation layer compared with the device bonded by Indium bumps. The deformation of the device bonded by Cu-Cu bumps with passivation layer is reduced by 19.5% to the device bonded by Indium bumps, and the decrease of the stress reached 59.1%. The great potential of Cu-Cu bonding structure with the passivation layer in the field of micro-display and micro-size packaging is demonstrated by the simulation results.

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